Audition & Speech Communication

Development of the Smart Speech™ serious game and the Smart Naso™ nasalance measurement apparatus for people with speech disorders (You et al., 2010 ~)

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Part II

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Learning Goals (1/2)

GOALS

- Evaluate the **characteristics and effectiveness of various warning and alarm signals** based on factors such as frequency, intensity, environmental conditions, and response requirements (simple vs. choice reaction).
- Describe **phonemes** and its characteristics such as **coarticulation** and **context dependency**.
- Analyze the differences in **acoustic power between vowels and consonants**, and their relative **impact on speech intelligibility**.
- Differentiate between the criteria of **speech intelligibility** and **speech quality** when evaluating the effectiveness of a speech communication system.

- Explain the **components of a speech communication system**.
- Describe the **acoustic characteristics of a speaker with high speech intelligibility**.
- Identify how **message characteristics**, such as vocabulary size, word length, and context, influence the intelligibility of speech in noisy environments.

Learning Goals (2/2)

GOALS

- Explain the **differential effects of transmission system** distortions, specifically noting why **center clipping** degrades intelligibility significantly more than **peak clipping**.
- Describe the effect of **reverberation** on speech intelligibility.
- Evaluate **speech intelligibility of a noisy environment** by calculating an **Articulation Index (AI)**.
- Identify the **factors of a listener** which affects speech intelligibility.

- Compare the fundamental mechanisms and the advantages/disadvantages of **"synthesis by analysis"** versus **"synthesis by rule"** in generating artificial speech.
- Apply human factors **guidelines for integrating synthesized speech into systems**, ensuring appropriate use for voice **warnings**, maximizing **intelligibility**, and considering **user acceptance**.

Section 05

Design of Auditory Displays for Specific Purposes

Guidelines for Warnings and Alarm Signals

Characteristics of Common Alarm & Warning Devices

Various types of audio devices each have their individual characteristics and corresponding advantages and limitations.

Alarm	Intensity	Frequency	Attention-getting ability	Noise-penetration ability
Diaphone (foghorn)	Very high	Very low	Good	Poor in low-frequency noise
Horn	High	Low to high	Good	Good
Whistle	High	Low to high	Good if intermittent	Good if frequency is properly chosen
Siren	High	Low to high	Very good if pitch rises and falls	Very good with rising and falling frequency
Bell	Medium	Medium to high	Good	Good in low-frequency noise
Buzzer	Low to medium	Low to medium	Good	Fair if spectrum is suited to background noise
Chimes & gong	Low to medium	Low to medium	Fair	Fair if spectrum is suited to background noise
Oscillator	Low to high	Medium to high	Good if intermittent	Good if frequency is properly chosen

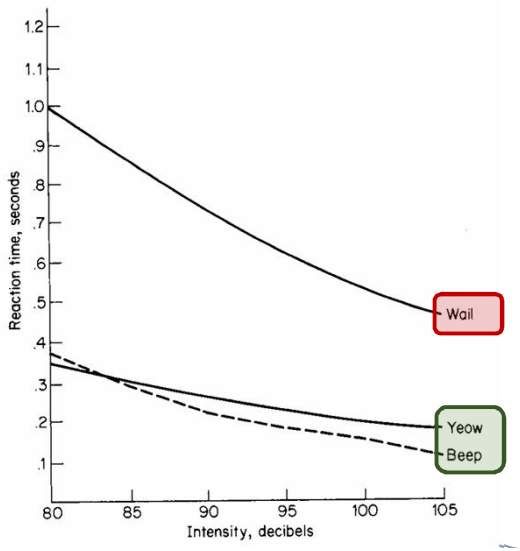
Design Recommendations for Alarm & Warning Signals

Frequency range	200–5,000 Hz general use; prefer **500–3,000 Hz (peak ear sensitivity)**.
Long distances	Use frequencies **below 1,000 Hz** (high frequency attenuates over distance > 300 m).
Around obstacles	Frequencies **below 500 Hz** bend around barriers and penetrate partitions.
Signal type	**Modulated signals** (1–8 beeps/s; warbling 1–3 Hz) command more attention than steady tones.
Spectral separation	Choose alarm frequencies in **bands where background noise is weakest**.
Multiple warnings	If different conditions require different signals, each must be distinctly **discriminable** and use **moderate intensity (85 ~ 95 dBA)**.
Dedicated channel	Use a communication channel (loudspeaker, horn) **NOT shared with other audio sources**.

Reaction Time to Alarm & Warning Signals

8 warning signals (beep, yeow, wail, yelp, whoop, intermittent horns, etc.) with various signal intensities tested across 5 ambient noise environments

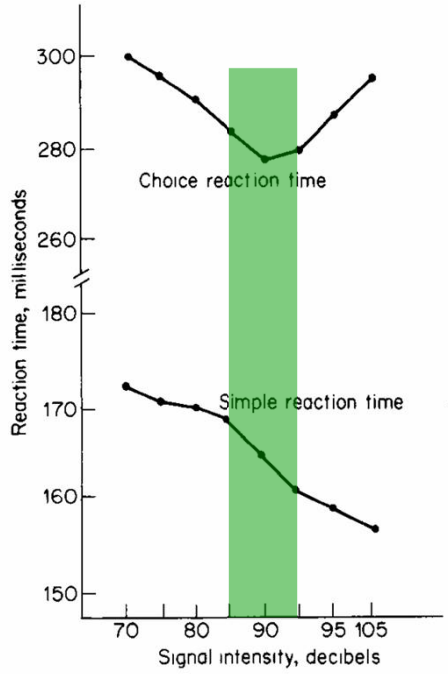
Beep (Best)	Intermittent horn: 425 Hz, on 0.7 s / off 0.6 s
Yeow (Best)	Descending frequency sweep: 800 → 100 Hz every 1.4 s
Wail (Worst)	Slow frequency rise 400 → 925 Hz over 3.8 s then descent



Simple vs. Choice Reaction Time

(Van der Molen & Keuss, 1979)

- **Simple RT**: **Higher intensity** → Faster response (monotonic improvement).
- **Choice RT**: **Moderate-intensity** (85 ~ 95 dBA) signals produced the FASTEST responses.



High-intensity startle reflex **DEGRADES choice performance** (response selection).

Section 06

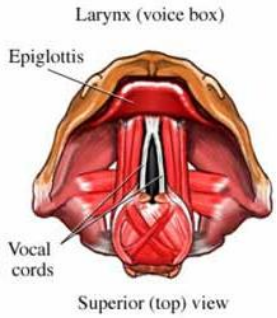
Nature & Evaluation of Speech

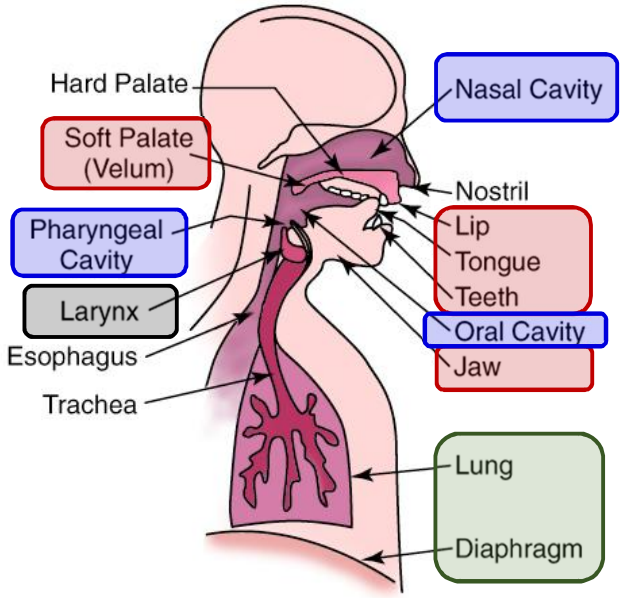
Speech production mechanics and speech evaluation criteria

Nature of Speech Production

From a HF/E perspective, speech is an **information display** — the source may be human OR machine.

Mechanics	Body Parts
Respiration	Lungs (breath stream source), diaphragm
Phonation	Larynx (vocal cords vibrate 80 – 400 Hz)
Resonance	Pharyngeal cavity, oral cavity, nasal cavity
Articulation	Soft palate (velum), tongue, teeth,, lips, jaw shape the spectrum



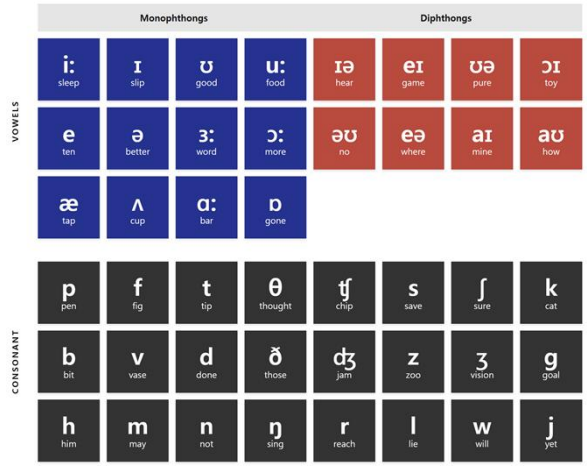


Phonemes

A phoneme is the **smallest unit of speech** that can change the **meaning of a word**.

Phonemic Chart

- English: 44 phonemes (20 vowels: 12 monophthongs + 8 monophthongs, 24 consonants)
- **Coarticulation**: The acoustic properties of a phoneme change with neighboring phonemes, yet we perceive them as the same. (e.g., the /d/ in 'di' vs. 'du', "Did you...?" [dɪdʒu])
- **Context Dependency**: Phoneme recognition is strongly context-dependent. In natural speech, ~50% of words are unintelligible when presented in isolation (Pollack & Pickett, 1964).
- **Vowels** carry most speech **POWER** (e.g., /a/ in 'talk' has 680× more power than /th/ in 'then' = 28 dB), while **consonants** carry most **INTELLIGIBILITY** information.

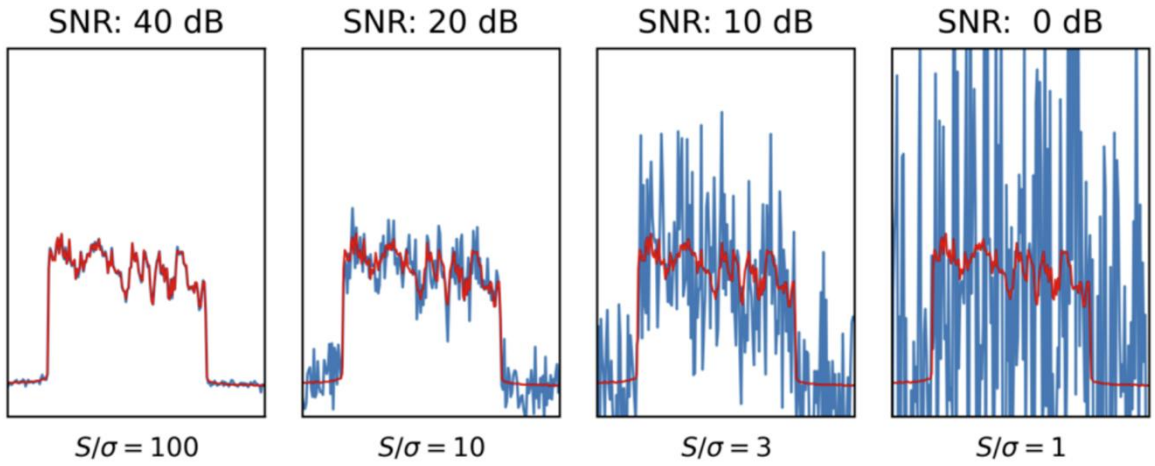


Vowels		Consonants
High acoustic power; carry the energy of speech		Low acoustic power; often 20-30 dB weaker
Provide volume but contribute less to meaning	VS	Carry the **CRITICAL cues** for intelligibility
More resistant to masking due to higher intensity		Easily masked by noise (esp. high frequencies)
Longer duration and simpler spectral structure		Design Implication: Protect 1–3 kHz band

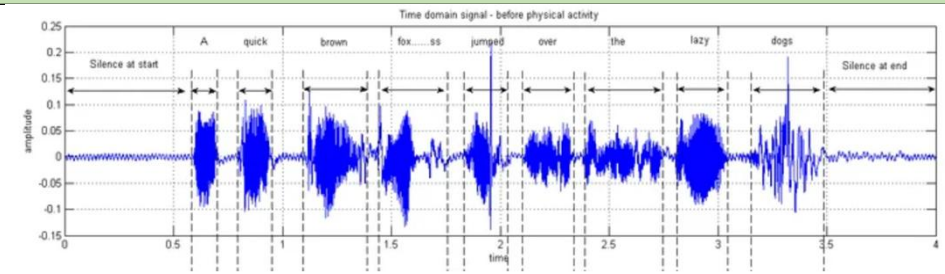
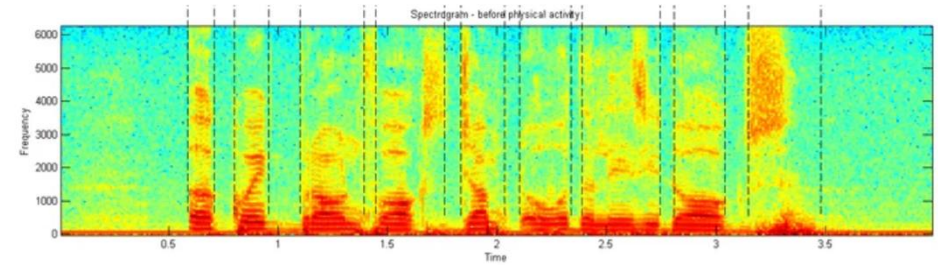
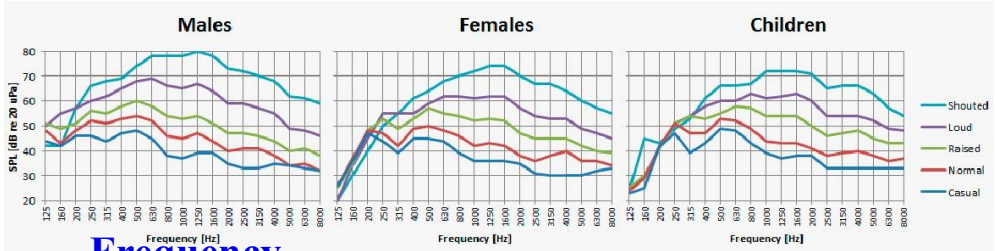
Speech Spectrum

- **Conversational levels: 55 ~ 65 dBA**; loud up to 85 dBA
- Intensity: **Males 3 ~ 5 dB louder** than females on average
- Frequency spectrum: **Males have more energy at lower frequencies** than females
- **Room acoustics and distance** reshape effective spectrum
- **Band-specific SNR (signal-to-noise ratio)** is decisive for comprehension

Intensity	
Voice level	**Approx. dBA**
Weak conversational	45
Normal conversational	**55**
Telephone/lecture	**65**
Loud/near-shouting	75
Shouting	**85**



Speech Visualization Methods

Type	Description	Illustration
Waveform	**Intensity over time**	**Intensity**  **time**
Spectrogram	**Frequency over time** with **intensity** shading	**Frequency**  **time**
Spectrum	**Intensity over frequency at a specific time**	**Intensity**  **@ time** **Frequency**

Speech Evaluation Criteria: Intelligibility and Quality

Speech Intelligibility

- Measure of message comprehension (% words recognized; accuracy)
- Assessed by transmitting speech to listeners who repeat or answer questions about what they heard
- Focuses on accuracy of information transfer
- Depends on consonants and context clues
- Can be high even with robotic/synthetic voices
- Critical for warnings, military, and safety systems

VS

Speech Quality

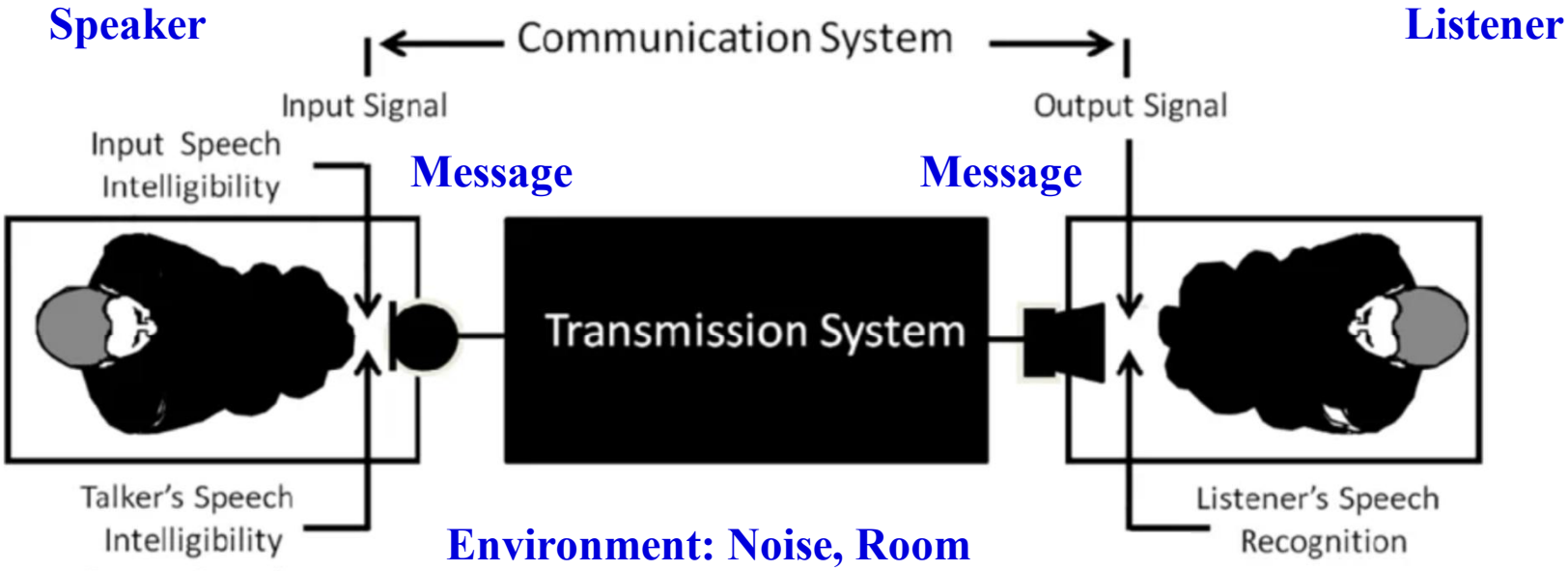
- Measure of naturalness and pleasantness
- Measured by preference ratings (excellent → unacceptable) or comparative preference tests.
- Focuses on speaker identity and realism
- Driven by vowels, prosody, and fidelity
- Essential for user satisfaction and long-term use
- Critical when speaker identity matters (phone calls)

Section 07

Factors Affecting Speech Intelligibility

Designing and analyzing a speech communication system for better speech indelibility

Speech Communication System



SI Factors: Speaker (1/2)

Speech Element	Description
Pronunciation	The act or result of producing speech sounds (phonemes) and word stress (syllable emphasis) in a specific way, often according to a recognized language standard or dialect.
Intonation	The rise and fall in the pitch of the voice during speech, used to convey meaning (such as a question vs. a statement), emotion, or sentence structure.
Rhythm & Timing	The pattern of stressed and unstressed syllables and the duration of sounds and pauses in speech, which together give a language its distinctive "beat" or flow.
Enunciation	The act of speaking clearly and distinctly; the precise articulation of individual speech sounds, syllables, and words so that they can be easily understood by the listener.
Speed (Rate)	The rate at which speech is delivered, typically measured in words per minute (WPM), which can influence comprehension, audience engagement, and perceived urgency.
Fluency	The ability to speak smoothly, easily, and readily, with minimal inappropriate hesitations, filled pauses (like "um" or "uh"), or disruptions to the continuous flow of thought.

SI Factors: Speaker (2/2)

Research (Bilger, Hanley & Steer, 1955): Superior speakers vs. less intelligible speakers differ on four acoustic characteristics:

Longer syllable duration	Allows more time for the listener's auditory system to integrate each phoneme.
Greater speech intensity	Higher energy level relative to ambient noise → better S/N ratio.
More speech time (fewer pauses)	Efficient use of transmission time; maintains listener attention.
Greater vocal frequency variation	Pitch variation improves rhythmic structure and attention modulation.

Speech training can yield **moderate improvements in articulation and intelligibility**, but deep structural articulator differences are largely fixed.

SI Factors: Message

Element	Description
Phoneme confusions	▪ MNDGBVZ, TKPFS
Word features	▪ Word-spelling alphabets (e.g., alpha, bravo) > singer letters (e.g., A, B) ▪ Familiar words > unfamiliar words ▪ Longer words > shorter words (provide more phonetic cues, aiding identification)
Context features	▪ Sentences > isolated words > nonsense syllables (context aids perception) ▪ Smaller vocabulary sets lead to higher intelligibility in noisy environments ▪ Use standard sentence constructions with information always transmitted in the same order (e.g., bin location, part number, part name, and quantity) ▪ Familiarise the receiver with the words and sentence structure to be used

Guidelines for an effective message

1. Avoid short words / single letters → use word-spelling alphabet (e.g., alpha, bravo, charlie)
2. Keep vocabulary small
3. Use standard sentence structure
4. Familiarize receiver with vocabulary and sentence structure

SI Factors: Transmission System

Filtering Effects

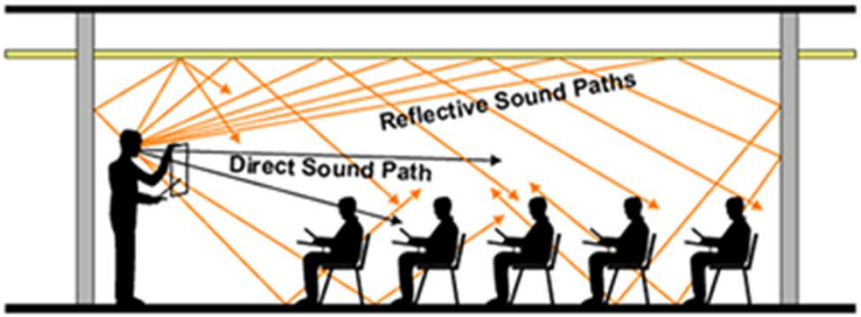
- Low-pass filter: cutting off above 4,000 Hz has little effect; cutting above 1,000 Hz severely degrades intelligibility.
- High-pass filter: cutting below 600 Hz has little effect; cutting below 3,000 Hz severely degrades intelligibility.
- Practical rule: frequencies in 1,000–4,000 Hz range are critical for intelligibility.

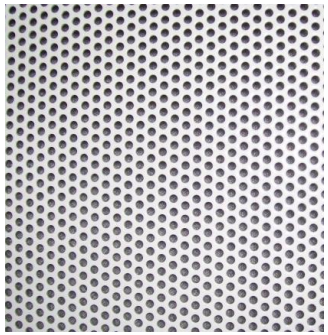
Peak Clipping (Amplitude Distortion)

- Clips peaks of vowel-dominated waveforms → reduces vowel power but leaves consonants intact.
- Up to ~6 dB clipping: speech still sounds normal.
- Clipping before adding to noise improves intelligibility over clipping after noise addition.
- Center clipping: Removing central amplitudes destroys consonants → Catastrophic for intelligibility.

SI Factors: Reverberation

- Definition: The **effect of sound bouncing back and forth** from the walls, ceiling, and floor.
- Measurement: The **time (denoted as RT_{60}) it takes for a sound to decay by 60 dB** after the source stops.
- Effects: **Intelligibility decreases linearly** with reverberation time (RT_{60})
- Guidelines: **RT_{60} = 1 ~ 2 sec. for concert halls**; **$RT_{60} \leq 0.5$ sec. speech-critical rooms**



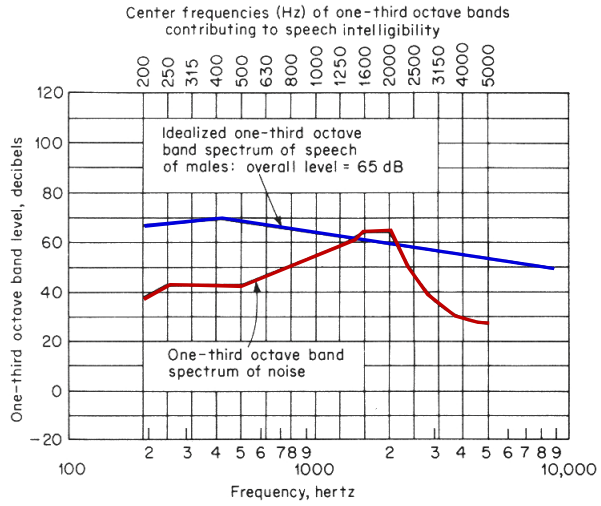


SI Factors: Noise (1/2)

The **Articulation Index (AI)** is a physical measure used to **predict the speech intelligibility** in a given **noise** environment.

Artificial index (AI) calculation procedure by ANSI

1	Plot one-third octave band levels of speech reaching the listener's ear on a worksheet.
2	Plot one-third octave band levels of steady-state noise reaching the listener's ear.
3	For each band: compute Speech Level – Noise Level. If noise > speech → assign 0. If difference > 30 dB → assign 30 dB.
4	Multiply each band difference by the band weight provided in ANSI tables. Record as Column 4.
5	Sum all Column 4 values → this sum is the AI.

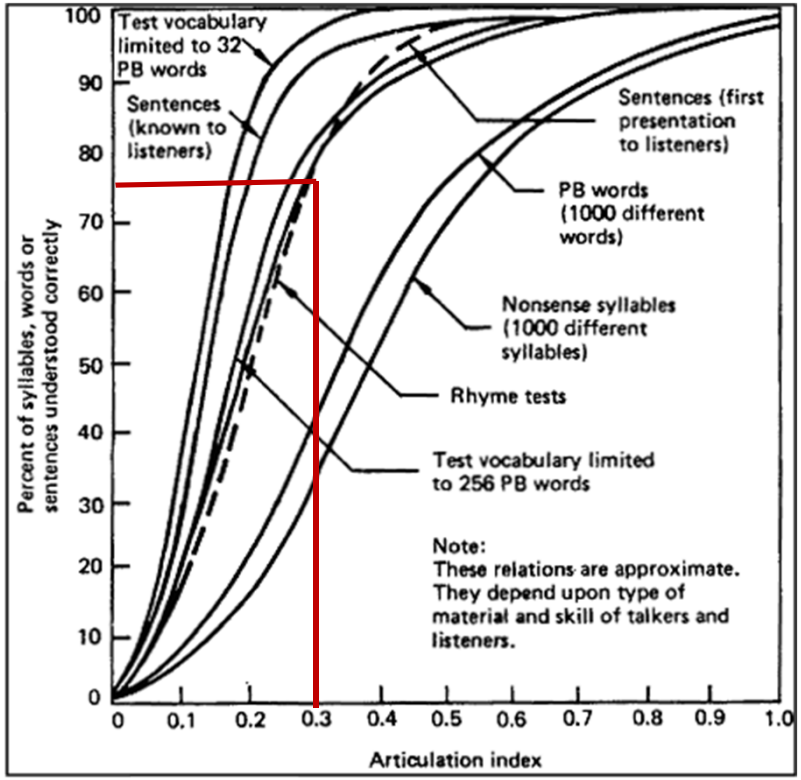


1. Band	2. Speech peaks minus noise, dB	3. Weight	4. Column 2 x 3
200	30	0.0004	0.0120
250	26	0.0010	0.0260
315	27	0.0010	0.0270
400	28	0.0014	0.0392
500	26	0.0014	0.0364
630	22	0.0020	0.0440
800	16	0.0020	0.0320
1000	8	0.0024	0.0192
1250	3	0.0030	0.0090
1600	0	0.0037	0.0000
2000	0	0.0038	0.0000
2500	12	0.0034	0.0408
3150	22	0.0034	0.0758
4000	26	0.0024	0.0624
5000	25	0.0020	0.0500
		AI =	0.4738

SI Factors: Noise (2/2)

Calculated AI Interpretation

AI	Sentence Intelligibility	Satisfaction Level
< 0.3	< 75%	Unsatisfactory to marginal
0.3 ~ 0.49	75% ~ 90%	Acceptable
0.5 ~ 0.7	90% ~ 99%	Good
> 0.7	~ 100%	Very good to excellent



SI Factor: Listener

Normal hearing	Required baseline for reliable speech reception. Hearing loss below 40 dB (mild) typically manageable; above 40 dB requires accommodation.
Age effects	Intelligibility decrements accelerate after age 60, especially for speeded speech (300 wpm), multi-talker backgrounds, and reverberant conditions (Bergman et al., 1976).
Hearing protectors: Broadcast	At noise > 80 dBA, protectors have little degrading effect and may enhance intelligibility. Below 80 dBA, protectors degrade broadcast speech.
Hearing protectors: Face-to-face	Both speaker AND listener wearing protectors → 15–40% degradation. In noise, people wearing protectors speak 2–4 dB softer and 20% faster.
Training	Listeners familiar with vocabulary, sentence structure, and speaker voice achieve significantly higher intelligibility, especially in noise.

Section 08

Synthesized Speech

Artificial Voice Generation & Human Interaction

Types of Speech Synthesis Systems

Types	Advantages	Disadvantages
Analog/Digital Recording	▪ Very high speech quality ▪ Natural-sounding voice ▪ Listener familiarity	▪ Fixed vocabulary — only pre-recorded words ▪ Mechanical parts wear (analog) ▪ Massive storage (digital): 8,000 samples/s × encoding = millions of bits for a small vocabulary
Synthesis by Analysis (Compressed Digitization)	▪ Less storage than straight digitization ▪ High quality comparable to recording ▪ LPC (linear predictive coding), waveform encoding	▪ Limited to pre-recorded vocabulary ▪ Unnatural coarticulation when words are concatenated (e.g., 'bookcase' = two /k/ sounds) ▪ Requires original speaker for vocabulary expansion
Synthesis by Rule (Text-to-Speech)	▪ Unlimited vocabulary — rule-based phoneme combination ▪ Very small memory footprint ▪ New vocabulary without human speaker	▪ Voice quality typically lower than digitized speech ▪ Prosody (rhythm, singsong) difficult to model perfectly

Performance with Synthesized Speech

Key factors governing whether an auditory signal will be detected:

99%	95%	84%
Natural speech	Best synthesized	Worst synthesized

Limited vocabulary, high S/N	Well-designed systems can achieve 99–100% intelligibility (Simpson et al., 1985).
Adaptation effect	With brief exposure or training, intelligibility increases significantly (Schwab, Nusbaum & Pisoni, 1985).
Talking signs	Intelligibility scores of 91–99% for familiar vocabulary in controlled conditions.
Memory burden	Synthesized speech requires MORE processing capacity → more trouble remembering messages vs. natural speech (Luce, Feustel & Pisoni, 1983).
Encoding solution	Increasing inter-word interval (1 s → 5 s) nearly doubled recall; pause-grouped digit strings improve recall (Waterworth, 1983).

Use Cases & Voice Selection

Application Scenarios

- **Aviation:** Machine-like warning voice preferred to avoid confusion with human co-pilot or ATC communications.
- **Consumer Devices:** Prioritize naturalness and likeability to enhance user experience (e.g., GPS, smart assistants).
- **Accessibility:** Clear articulation with adjustable speech rate and pauses for visually impaired users.
- **Industrial Control:** Concise phrasing and confirmable alerts to ensure safety in high-noise environments.

Context Matters

Voice personality should match the urgency and context of the task.

User Acceptance

Natural sounding voices increase acceptance in non-critical systems.

Safety First

Distinctiveness prevents masking and confusion during emergencies.

User Preference for Synthesized Speech

Intelligibility ≠ Preference. A system can be intelligible yet unpleasant — or pleasant yet incomprehensible.

Market Rejection	A microwave oven with speech instructions saw such poor sales the manufacturer stopped production. ATM users disliked voice messages when others were present.
Perceived Quality Descriptors	Synthesized speech commonly described as: machinelike, choppy, harsh/grainy, flat, noisy. Lack of coarticulation (word blending) and natural intonation are primary complaints.
Context-Dependent Preference	Cockpit pilots often PREFER a machine-sounding voice for warnings — to clearly distinguish from human crew voices.
Intonation Pattern Matters	Continuant (rising pitch) group delivery of phone numbers preferred over neutral/terminator. Subtle prosodic cues significantly affect user preference.

Guidelines for the Use of Synthesized Speech

1

Voice differentiation

Warnings should use a qualitatively different voice from other voices in the system — especially cockpit environments.

2

Alerting tones

If synthesized speech is used EXCLUSIVELY for warnings → no preceding alerting tone needed. If used for other info too → add a warning-specific tone cue.

3

Maximize intelligibility

Select or tune the system for highest intelligibility, especially for safety-critical messages.

4

Natural voice for general use

For general-purpose information (not warnings), maximize naturalness to improve user acceptance.

5

Replay mode

Provide replay capability, especially for complex information. (But note: users often do not actually use it — Roelofs, 1987.)

6

User interrupt

Allow experienced users to interrupt the message mid-stream. Critical for efficiency.

7

Training period

Provide introductory exposure to the system's voice before operational use.

8

Use sparingly

Avoid 'high-tech fever' — use only where the auditory modality genuinely serves users better than visual alternatives.